day-03 Short Question Assignment

Question 1: The layout Divide

you are taking the derivative of a loss Scalar L with respect to a weight matrix W that has Dimension 256X128.

If you see the denominator layout, what will the exact dimension of the resulting derivative matrix be?

Solution: The gradient matrix 256X128, Denominator layout preserves the exact shape of the matrix you are differentiating with respect to.

Question 2: Gradient routers.

the upstream gradient arrives at an addition Node $Z=(x+y)$

CamScanner

If the upstream gradient is exactly 4.5, what value is passed backward along the x branch, and what value is passed backward along the y branch?

Solution:

Both x and y receive exactly 4.5, The local derivative of x+y with respect to either variable 1. The chain rule is $4.5 \times 1 = 4.5$

Question 3: Gradient switchers

During Backward pass, the upstream gradient arrive at multiplication node $(z = x.y)$.

During the forward pass, the cached inputs were $x=2$ and $y=-3$.

If i/p stream gradient is 10, what is the final downstream gradient passes backward to the x node

Solution:

The downstream gradient ~~for~~ x is -30. The up stream gradient 10 is multiplied by local gradi (the value of y, which is -3).

Question 4: Forward pass caching

What exactly is being stored in memory during this forward pass, and why is it mathmatically required before the backpass can even begin?

Solution:

The intermediate action value (the numerical results of every mathmatical operation). They are required because local derivates often depends on inputs.

Question 5: Analytical vs Numerical.

You are asked to compute the gradients for

a Neural Network with 10 million parameters.

<u>True / False</u>: Approximate the gradients numerically using the limit defination ($h = 10^{-5}$) is computationally faster than calculating the analytical gradients using Backpropagation Becase it avoids building a complex computational graph in memory.

<u>Solution</u>: <u>False</u>: Slower, requires running a full Forward Pass for every single parameter in the network.

CS CamScanner